



RBrow 2

Dalhousie University

Department of Psychology

Twentieth Annual
In-House Convention

28-29 April 1994

SCHEDULE

Thursday, April 28, 1994

9:30 - 9:45 Coffee

9:45 - 10:00 Opening Remarks—Richard Brown, Chairman

10:00 - 11:30 Session 1. Chair: Brad Frankland

- T1) R. Hoffman & E. Boychuck
- T2) B. Losier
- T3) M. MacGregor & K. Davidson
- T4) R. Klein, A. MacDonald & J. Christie
- T5) J. Christie & R. Klein

11:30 - 1:30 Lunch (Posters should be ready by this time)

1:30 - 2:30 Session 2. Chair: Heather Schellinck

- T6) K. Forbes & C.C. Mate-Kole
- T7) M.G. Yoon & M. Yoon
- T8) D.E. Mitchell, G. Bateman, C.J. Beaver & P. Ritchie
- T9) J. Hamm & Tracey Taylor

2:30 - 3:00 Afternoon Tea

3:00 - 4:00 Session 3. Chair: Dennis Phillips

- T10) J. Ourednik & W. Ourednik
- poster T11) F. Dastur
- T12) S. Bishop
- T13) M. Antle, H. Piggins & B. Rusak
- T14) D.P. Phillips, M.N. Semple & L.M. Kitzes

4:00 - 6:00 Posters (Lounge):

- P1) H. Schellinck & R. Brown
- P2) J.L. Ingles, C.C. Mate-Kole, and J.F. Connolly
- P3) S. Morehouse & J. Fisk
- poster P4) P.A. McMullen & T. Trappenberg
- P5) G. Rioux
- P6) L.A. Chambers & K. Davidson

There will be wine and cheese served during the poster session.

Friday, April 29, 1994

9:45 - 10:00 Coffee

10:00 - 11:30 Session 4. Chair: *Hugh Piggins*

- T15) B.W. Frankland & P.A. McMullen
- T16) W.K. Honig, W.R. Matheson & S. Gabois (Speaker)
- T17) S.L. Wright
- T18) M. Twum & G. Eskes
- T19) D. Cantoni

11:30 - 1:30 Lunch (Posters should be ready by this time)

1:30 - 2:30 Session 5. Chair: *George Rieck*

- T20) G. Scott & B. Rusak
- T21) A.-M. Brown
- T22) H. Piggins, D. Cutler, K. Semba & B. Rusak
- T23) S.H. Stewart & S.B. Zeitlin

2:30 - 3:00 Afternoon Tea *see Theme*

3:00 - 4:00 Session 6. Chair: *Sherry Stewart*

- T24) T. Taylor and R. Klein
- T25) R. Brown, T. Calder & C. Gilbert
- T26) K. Semba
- T27) J. Barresi

4:00 - 6:00 Posters (Lounge):

- P7) A. Trillo et.al.
- P8) F.N. Dastur & R.E. Brown
- P9) T.L. Fidler & V.M. LoLordo
- P10) J. Francis
- P11) M.W. MacGregor & K. Davidson
- P12) S.D. Sweet, G. Rioux & P.J. McGrath
- P13) C.A. Dywan, J. McGlone & A. Fox
- P14) *There's a list*

There will be pizza and beer served during the poster session.

Awards will be presented during the Poster Session on both days.

T1. Ron Hoffman, Psychology & Elaine Boychuk, Assoc. Univ. Librarian, Access Systems

LAB, LIBRARY, LAN, AND (L')INTERNET

This past year has seen the arrival of some exciting new technologies for information access. At Dalhousie, the University Library has been purchasing multi-user network versions of several computer-readable databases, including those most widely used by students in Psychology. It is now possible for our students to access CD-ROM databases located in the Killam Library from any terminal in the Psychology Computer Laboratory. At the same time (and from the same computer), students can check Novanet for availability of journals at the libraries, and even order a fax copy of an article over the Internet. For the benefit of those who have not yet utilized these new tools, we will quickly demonstrate use of PsycLIT, MedLine and Science Citation Index, and then introduce some new tools for finding information on the Internet.

Veronica

T2. Bruno J. Losier

LINE BISECTION AND VISUAL ORIENTATION ERRORS ALONG HORIZONTAL AND VERTICAL PLANES IN PERIPERSONAL AND EXTRAPERSONAL SPACE IN PATIENTS WITH VISUOSPATIAL NEGLECT: A PROPOSAL

In recent years, line bisection and visual orientation studies of visuospatial neglect have been prolific, but the evidence provided has been fixed by physical limitations and methodological moderations. Line bisections are performed manually and are confined to peripersonal space. Thus, knowledge of visuospatial deficits outside of arm's reach (e.g., extrapersonal space) is missing. Similarly, the visual orientation task is carried out by comparing responses along a left-right continuum. Thus, investigation of visuospatial neglect along multiple planes is absent. The purpose of the present study will be twofold. First, it proposes to investigate line bisection and visual orientation errors along horizontal and vertical planes in peripersonal and extrapersonal space. Second, a model explicating line bisection errors is missing. It is suggested that the theoretical framework of the visual orientation task may provide an opportunity to further elucidate on mechanisms of line bisection errors. Thus, the present study intends to illuminate mechanisms underlying bisection errors by comparing displacement errors on line bisection to errors on a visual orientation task. A proposal outlining the background and research plan will be presented.

- T3. Michael MacGregor & Karina Davidson

GENDER DIFFERENCES IN THE RELATION BETWEEN MEASURES OF HOSTILITY AND RESTING SYSTOLIC BLOOD PRESSURE

Gender differences in the ability of the Cook-Medley Hostility Scale (HO) and its subscales, and the Anger Expression Scale (AES) to predict resting systolic blood pressure were examined. Four blood pressure readings were taken at three-minute intervals and the fourth reading was used as the baseline. Following baseline readings, 99 male and 109 female undergraduate subjects completed the HO and the AES.

Results indicated that for males the following HO scales were significantly positively correlated with resting systolic blood pressure: The HO total hostility score ($r = .245, p < .01$), Costa's subscales of cynicism and paranoid alienation ($r = .231, p < .05$ and $r = .263, p < .01$ respectively), and Barefoots' subscales of cynicism, hostile emotion, and aggressive responding ($r = .208, p < .05$; $r = .328, p < .001$; and $r = .315, p < .001$ respectively). For females none of the HO subscales were significantly correlated with resting systolic blood pressure. There was no relation between the AES and resting systolic blood pressure for either gender.

These results are discussed in terms of the implications of using traditional measures of hostility for predicting risk for CHD in male and female populations.

- T4. Raymond Klein, Alan MacDonald & John Christie

HOW CAN WE DETERMINE IF ATTENTION IS THE GENERATOR RESPONSIBLE FOR ILLUSORY LINE DRAWING?

Two circles, separated by 4° are displayed on a screen. One briefly brightens and then a line connecting the two circles is added to the display. Although the line appears instantaneously, the subject's subjective experience is that the line is "drawn" from the flashed circle toward the unchanged one. The flashed circle does attract the observer's attention, and Shimojo, who discovered this line drawing illusion, attributes it to the locus of attention. Alternatively, the illusion might depend on purely visual mechanisms activated by the stimulus, such as those associated with apparent motion. We plan to compare the strength of the illusion under conditions of exogenous (externally generated, e.g. following abrupt peripheral luminance increment) and endogenous (internally generated, e.g. following verbal instruction) control over attention. Our first step was to determine if the line drawing illusion could be cancelled by presenting a line that WAS drawn over time in the direction opposite to the illusory motion.

- T5. John Christie & Raymond Klein

NEGATIVE PRIMING: OBJECT FILE REVIEW OR SELECTIVE INHIBITION? THE METHOD MATTERS!

Negative priming is the impaired performance that occurs when a subject has to respond to a target that occupies the same space as a previously ignored distractor. Object file review and selective inhibition are the two principal explanations for this phenomena. Object file review explains negative priming as caused by mismatch between the attributes of the target and those of the item that previously occupied the location. According to selective inhibition negative priming occurs because a distractor is not merely ignored, but actively inhibited as part of the selective process and that this carries forward to items presented later in the same space. Two experiments, using the most common methodologies were designed to test these two theories. It was also found that the relative contribution of object file review and selective inhibition depends on, as yet indeterminate methodological factors.

- T6. Kelly Forbes & C. Charles Mate-Kole

THE NART AS A TOOL FOR ESTIMATING PREMORBID INTELLIGENCE IN A NORTH AMERICAN POPULATION: A WILD GOOSE CHASE?

Current level of cognitive performance is not indicative of premorbid level of functioning. Estimates of premorbid intelligence are essential for determining the extent of cognitive impairment in brain damaged populations. A relatively new approach to estimating premorbid intelligence is based upon the relationship that exists between word-reading ability and general intelligence (Nelson & McKenna, 1975; Nelson & O'Connell, 1978). The National Adult Reading Test (NART), which contains orthographically irregular words, was normed on a British population and provides estimates based upon the WAIS. The present study based estimates of intelligence in a North American population on the WAIS-R. NART based predictions of intelligence were compared to other estimates based upon the Vocabulary Age Scaled score and the VIQ-PIQ discrepancy in performance on the WAIS-R. Results indicate that the NART is a useful tool for predicting premorbid intelligence in normal and brain damaged populations within a North American population.

- T7. Myong G. Yoon & Myonghae Yoon

CHILDREN'S UNDERSTANDING OF MENTAL MISREPRESENTATIONS

We studied the development of children's understanding of two common types of mental misrepresentations: Holding false beliefs and forgetting previous beliefs. A person may hold a false belief about a current state of affairs due to ignorance of externally caused changes or perceptual illusions. Or a person may misrepresent a current state of affairs by simply forgetting its previous mental representation caused by internal changes within the person. The results of our experiments suggest that children's abilities for retroactive inferencing of the protagonist's forgetting from her incongruous action begin to emerge around 4y8m, about a half year later than their abilities for attribution of a false belief.

- T8. D.E. Mitchell, G. Bateman, C.J. Beaver, and P. Ritchie

SINGLE-CELL RESOLUTION OF THE CHANGES IN THE PATTERN OF INNERVATION OF AREA 17 BY THE TWO EYES FOLLOWING MONOCULAR DEPRIVATION AND ITS REVERSAL IN KITTENS

C-Fos, the protein product of the c-fos gene, has been shown ⁽¹⁾ to be expressed rapidly in the visual cortex of kittens following brief exposure to light. Our purpose was to explore the use of Fos immunohistochemistry as a means for examining the three dimensional pattern of innervation of area 17 by the two eyes in monocularly deprived kittens as well as the speed and nature of the changes that occur subsequently following reverse occlusion. The experiments were performed on kittens that had been monocularly deprived at 1 week of age until they were 5 weeks old. To visualize the pattern of innervation of area 17 by a given eye, the animal was first placed in a darkroom for a period (24 hrs) that was sufficient for constitutive Fos expression to be driven below baseline levels of detectability. The kitten was then allowed normal visual experience with just the eye of interest open for 2 hours at which point the animal was euthanized and perfused. The immunocytochemical procedures ⁽¹⁾ were conducted on 50 um sections cut from unfolded and flattened visual cortex.

In animals exposed through the deprived eye, immunoreactive neurons were observed in patches in superficial and deeper layers of area 17. The number of immunoreactive neurons within each patch as well as the size and shape of the patches changed with increasing length of deprivation. This method permits, for the first time, visualization with cellular resolution of the columnar arrangement of territory innervated by a given eye throughout area 17 and as such may prove to be a superior method for assaying the effects of deprivation than existing methods.

⁽¹⁾ Beaver et al. J. Comp. Neurol. 333:469-484, 1993.

- T9. Jeff Hamm (ENTP) & Tracy Taylor (INTP)

THE FAST LETTER GAME

When subjects identify a target in a rapid serial visual presentation (RSVP) stream, their ability to detect a subsequent probe stimulus is reduced. Known as an attentional blink (AB), we determined whether this phenomenon depends on shared categorical membership between target and probe stimuli. We presented subjects with RSVP streams of black letters in which one white (target) letter was embedded. In half the streams, a probe (O) was one of eight post-target stimuli. For half the subjects this probe was referred to as the letter "O"; for the other half it was referred to as the digit "zero". Relative to an ignored-target control, an AB occurred in both experimental groups. However, the AB was attenuated when subjects detected zero rather than "O". This dependence of AB on categorical membership of target and probe stimuli was confirmed in a study that had subjects detect a "6" following letter identification.

- T10. Jitka Ourednik, Psychology & Wenzel Ourednik, Anatomy and Neurobiology

EMBRYONIC NEURAL GRAFTS INDUCE REPAIR BY THE JUVENILE NEOCORTEX IN MICE - AN IMMUNOHISTOCHEMICAL AND AUTORADIOGRAPHIC STUDY

The postnatal central nervous system (CNS) of mammals has a diminished restorative capacity. Recent research proposes that embryonic neural grafts transferred into a damaged region of the CNS may lead to a certain degree of anatomical and functional recovery. Until now, it was generally believed that it is the graft which is the source of the reconstituted tissue. ***Our observations*** (J.Ourednik et al. 1993, W. Ourednik et al. 1994) ***indicate that in the CNS the regenerative capacity is not lost irreversibly and that embryonic grafts from the presumptive murine cerebral cortex may induce restorative processes and cell proliferation within the severely injured cerebral cortex of young mice.***

Our investigation is based on immunohistochemical distinction of donor and host tissues of two congenic mouse strains which express two different allelic forms of the Thy-1 antigen, and on detection of newly formed cells by their incorporation of ³H-thymidine administered to the recipients shortly after transplantation. Neurons are revealed by antibodies against neuron-specific marker proteins (MAP2, PV or NF-h). Radioactively labelled cells are detected in the immunohistochemically stained sections by autoradiography.

We demonstrate that in animals that had received a tiny graft, a serious mechanical insult of the neocortex was repaired mainly by proliferation of host cells of both neuronal and non-neuronal origin. During the 2-month survival period, the graft was either completely resorbed, or participated in a minor way in the reconstruction.

Ourednik, J., Ourednik, W., Van der Loos, H., Do fetal neural grafts induce repair by the injured juvenile neocortex? *NeuroReport* 5: 133-136 (1993)

Ourednik, W., Ourednik, J., the late Van der Loos, H., Newly formed host cells in grafted juvenile neocortex expressing neuron-specific marker proteins. *NeuroReport* 5: (1994, in press)

- T11. Farhad N. Dastur

OLFACTORY AND TASTE ABILITIES IN PRE- AND POSTPARTUM WOMEN

This study will assess the olfactory and taste abilities of women before and after birth. Sense of smell and taste may be altered during and after pregnancy for three reasons: (1) smell and taste are sensitive to endocrine changes and pregnancy involves significant endocrine changes (e.g. progesterone, estrogen, HPL, HCG); (2) pregnancy-associated alterations in food palatability may be an adaptation to protect the embryo from food toxins, and these changes may be mediated by smell and taste changes; and (3) olfaction may play an important role in the mother's ability to recognize and bond with her infant, and thus may be enhanced postpartum. Subjects' performance will be measured in a longitudinal design once during the 2nd trimester, once shortly after parturition (~2 weeks), and finally at 3 months postpartum. Smell will be assessed using an odour identification test and an odour threshold detection test. Taste will be assessed using a whole-mouth taste recognition test. Theoretical and methodological criticisms are welcome.

T12. Scott Bishop

**DOES ATTENTION MEDIATE THE RELATION BETWEEN
CATASTROPHIZING AND PAIN?**

Catastrophizing has been identified as an individual difference variable that contributes to increased distress reactions during painful experiences. Catastrophizing has been described as a tendency to engage in pain-related negative ideation characterized by rumination, magnification of threat, and lack of confidence and control. Research has been consistent in showing that catastrophizing is associated with increased pain and emotional distress during painful stimulation.

However, there has been little research examining the mechanisms that mediate the relation between catastrophizing and distress reactions. Based on recent findings in our lab, I will present a model in which will suggest that attentional processes mediate the relation between catastrophizing and pain. Future research and implications will also be discussed.

T13. Michael Antle, Hugh Piggins, & Benjamin Rusak

Getting a GRiP on GRouP GRoPe in the Light

In rodents, the suprachiasmatic nucleus (SCN) of the hypothalamus is the circadian pacemaker. The SCN contains various substances which may be involved in photic entrainment of the pacemaker. The bombesin like peptide, gastrin-releasing peptide (GRP) is one such substance. The present study was designed to determine if GRP would phase shift hamsters housed in continuous illumination. GRP was found to produce "light pulse-like" shifts when injected to the SCN. These results suggest that GRP may be involved in photic entrainment.

T14. D.P. Phillips, M.N. Semple and L.M. Kitzes

**FUNCTIONAL SPECIALIZATION IN THE CAT'S AUDITORY CORTEX
(Or: What Needles in the Posterior Area Reveal About God's
Inventiveness)**

The cat's auditory sensory cortex contains a number of territories distinguished by their anatomy and the internal organization of their frequency maps. Much quantitative evidence is available on the physiology of cells in the primary auditory field (AI), but virtually nothing is known of the physiology of neurons in the other territories. In anesthetized, adult cats, we explored the posterior auditory field (P), studying individual neurons with stimulus paradigms that have previously been used to study AI cells. We found that the majority (70-90%) of field P neurons had properties seen in a minority (20-30%) of AI cells: powerful lateral inhibition, apparent tuning to the amplitude of preferred-frequency tones, and sensitivity to the rise-time of tonal signals. The minimal latencies of field P cells were slightly longer than those of AI cells. These findings raise the question of whether field P cells derive their input from the subset of AI cells that already possess these properties.

- T15. B. W. Frankland (INTP) & P. A. McMullen

WeOn Geons

Biederman's (1987) Recognition-by-Components proposal argued that a limited set of categorically-perceived, geometric, three-dimensional visual primitives (geons) form the (neurological) basis of all object perception. To test the prediction of categorical perception, a continuum of three-dimensional volumes between a block and cylinder (two geons advocated by Biederman) was constructed, rendered in a two-dimensional computer display, and presented to subjects. In a multiphase, two-alternative, timed, forced-choice decision, subjects repeatedly labelled each volume as block or cylinder. In Phase 1, subjects established a baseline boundary between the block and cylinder. The sharp discontinuity between the two provided evidence for a categorical boundary. In Phase 2, subjects were presented with either the block or cylinder in an attempt to exhaust the (neurological) detectors for that object (adaptation). In Phase 3, the categorical perception of block versus cylinder was reassessed. After adaptation, relative to its Phase 1 position, the categorical boundary shifted towards the adapting stimulus. The shift was in the predicted direction supporting the existence of geons.

- T16. W. K. Honig, William R. Matheson & Simon Gadbois

COMPARISON OF ABSOLUTE AND RELATIVE NUMEROSITY DISCRIMINATION BY THE PIGEON

Two groups of six pigeons were trained in an operant behavior chamber to peck at a video screen on which arrays of either red or blue "small" squares (1 cm on each side) or "large" squares (2 cm on each side) were presented in a 3 X 3 matrix. They had to peck differentially at the left or right side of the screen on the basis of the absolute numbers of squares or their relative proportions. In the relative discrimination, one side of the screen was positive when 0 to 4 red squares were presented together with 9 to 5 blue squares, while the other side was correct for the opposite proportions. In the absolute discrimination, one color was omitted, so that two sub-groups were trained, only with blue or with red squares, to peck at one side of the screen for 0 to 4 squares, and at the other side for 5 to 9. Large and small squares were presented to control for total areas of squares. Both groups provided orderly discrimination gradients. The relative gradients were quite linear, and were steeper when large squares were presented. The absolute gradients decreased in slope with larger numbers of squares, and, for a given number of squares, the pigeons pecked more to the side correct for "many" when the squares were large rather than small, suggesting that there was an effect of area as well as numerosity.

T17. Sandra L. Wright

THE IMPORTANCE OF PATERNAL CARE FOR PUP SURVIVAL AND DEVELOPMENT IN *MUS MUSCULUS*.

Studies of parental behaviour in rodents generally provide ad lib food and water in a warm environment and under these conditions the presence of the male has little or no effect on pup survival and development. We examined paternal care in a polygynous rodent (*Mus musculus*) when wheel running was necessary to obtain food. Single mothers (Father Absent) and paired mothers (Father Present) were compared under three different foraging requirements. Paired mothers weaned significantly more pups than single mothers when wheel running was required for obtaining food (Wheel Contingent). There were no significant differences of the father's presence on pup survival in the No Wheel (standard laboratory housing with ad lib food) or Wheel Noncontingent groups (running wheel with ad lib food). Pup weight was lower in the Wheel Contingent group than in the other two groups and the father's presence had no significant effect on pup weights. Both single and paired mothers showed parental care in the light phase and wheel running in the dark phase of the LD cycle. However, paired mothers alternated parental care and running with the fathers. Fathers directly facilitated pup survival by spending as much time in parental care as mothers and indirectly by running on the wheel to earn more food than they consumed. These results suggest that paternal care benefits pup survival more in naturalistic conditions where the parents must forage to obtain food than in the standard laboratory environment where food is provided ad lib.

T18. Maxwell Twum & Gail Eskes

NOW YOU SEE IT, NOW YOU DON'T: ROLE OF ATTENTION IN PRIMING

Many recent studies show that attention manipulation has little effect on perceptually driven implicit tests of memory. In these studies, priming (nonconscious facilitation of processing due to previous study) is obtained even when attention is divided between the target information and another ongoing task. There are, however, no studies that have directly compared the amount of priming obtained as a function of the amount of attention given to target information. The present study was conducted to further investigate whether focal attention was necessary for priming. It was hypothesized that if focal attention was crucial for priming, then greater priming could be expected where attention was focused on the target information. The Stroop Color-Word paradigm was used to manipulate attention to the words. In this paradigm, different groups of subjects either read words, named the ink in which the the words were printed or named the ink color of a series of XXXs which were presented simultaneously at different locations on a computer screen. Retention of the words was later tested using both cued recall (explicit) and word fragment completion (implicit) tests. Priming was obtained in all conditions of the experiment. However, only priming for the word-reading condition was significantly different from baseline performance. There were no differences in priming for naming the ink of the words and the XXXs. There was also greater explicit cued recall when subjects read the words. Implication of these results will be discussed in the framework of attention and information processing.

T19. Debora Cantoni

SOCIAL AND SPATIAL ORGANIZATION OF FREE-RANGING SHREWS

The social organization of free-ranging *Sorex coronatus* and *Neomys fodiens* was studied by measuring the spatial distribution and temporal avoidance or aggregation of individuals, using simultaneous radio-isotope tracking of several shrews. *Sorex coronatus* was strictly territorial during the non-breeding season, with little overlap between home ranges and low coincident activity between individuals. In the breeding season, females remained strictly territorial, whereas large home range overlap between males and females and also between males was observed. During this season, males kept intensive watch on one female and occasionally visited a second, in turn was regularly visited by another male. This behaviour suggests a monogamous mating system with a tendency towards polygamy. *Neomys fodiens* was territorial during the non-breeding season. Each individual defended a portion of a river but a slightly greater home range overlap was observed between than within the sexes. During the breeding season, females remained strictly territorial, whereas males tended to wander, indicating an ~~overlap~~ promiscuous mating system. After weaning, juveniles of both species dispersed rapidly and became territorial until the following breeding season. These results were compared with published data on *Crocodyrus russula*. During the non-breeding season, there are two types of social organization: a colonial system in *C. russula*, where coincident rest between individuals allows them to reduce their daily expenditure of energy by social thermoregulation; and a territorial system in *S. coronatus* and *N. fodiens*, which may ensure an adequate food supply. During the breeding season for all three species females are strictly territorial perhaps to ensure they have sufficient food for their litters. For the males, three different strategies have been observed and these are also discussed.

T20. George Scott & Benjamin Rusak

METABOTROPIC GLUTAMATE RECEPTORS IN THE HAMSTER SUPRACHIASMATIC NUCLEUS.

The suprachiasmatic nuclei (SCN) of the mammalian hypothalamus function as a pacemaker which controls mammalian circadian rhythms. The SCN receive photic input via the geniculohypothalamic tract (GHT), an indirect visual pathway arising from cells in the intergeniculate leaflet (IGL) of the lateral geniculate body, with neuropeptide Y and GABA being likely neurotransmitter candidates. The SCN also receive direct visual input via the retinohypothalamic tract (RHT), and evidence suggests that the excitatory amino acid glutamate mediates neurotransmission in the RHT. Recent reports have demonstrated the presence of ionotropic excitatory amino acid receptors of the NMDA and non-NMDA receptor subclasses in the SCN. In addition, metabotropic excitatory amino acid receptors have been located in the SCN. To date, very little information is available on the effects of metabotropic receptor activation in the SCN, however, initial studies in our laboratory have demonstrated that 1S,3R ACPD, a specific metabotropic receptor agonist, causes a dose-dependent increase in neuronal activity in the SCN *in vitro*. Further studies aim to pharmacologically characterise metabotropic glutamate receptors in the SCN and elucidate the role of this receptor type in the SCN *in vivo*.

T21. Anne-Marie Brown

MEMORY FUNCTIONS IN MULTIPLE CHEMICAL SENSITIVITY

Case series of patients with Multiple Chemical Sensitivity (MCS) suggest difficulty on several memory tests in addition to cacosmia. Bell (1992) suggested that this pattern of symptoms might implicate focal dysfunction of the temporal lobes. To test this hypothesis a case control study was initiated to compare the pattern of memory loss in MCS, healthy controls matched for age, education and sex and patients with temporal lobectomy on measures of estimated pre-morbid ability, intelligence, attention, immediate and delayed memory for verbal and nonverbal material. Twenty-six employees from a "sick building" were referred for extensive neuro-psychological evaluations based on impairments documented in cognitive screening. The majority had MCS. Group results for MCS patients, controls and patients with temporal lobe lesions will be presented.

T22. Hugh D. Piggins, David J. Cutler, Kazue Semba & Benjamin Rusak

TIME FOR SUBSTANTIAL P?

The tachykinin substance P (SP) is found throughout the mammalian central nervous system. We have demonstrated SP-like immunoreactivity in cells, fibres, and terminals of the Syrian hamster suprachiasmatic nuclei (SCN), which contain the primary circadian pacemaker. The possible role of SP in circadian processes has yet to be established. We examined the effects of ionophoretically applied SP on the extracellularly recorded firing rates of hamster SCN cells maintained in an *in vitro* brain slice preparation. SP evoked responses from ~54% of 85 cells tested. Most responsive cells were activated (~85%) while ~15% were suppressed by SP. A greater proportion of cells responded when tested during the projected day (~72%) than during the projected night (~38%). These preliminary data indicate that SP can alter hamster SCN neuronal activity, particularly during the subjective day.

T23. Sherry H. Stewart & Sharon B. Zeitlin, Psychology, University of Toronto

UNIVERSITY STUDENTS' DRINKING MOTIVES: RELATIONSHIPS TO ANXIETY PRONENESS, ANXIETY-SENSITIVITY, AND GENDER

The present study examined the relationships between anxiety-proneness (trait anxiety), anxiety-sensitivity (fear of anxiety), and self-reported reasons for drinking using a newly-validated measure of drinking motives. The State-Trait Anxiety Inventory-Trait subscale (STAI-T), the Anxiety Sensitivity Index (ASI), and the Drinking Motives Questionnaire (DMQ) were administered to 314 university students. Higher ASI scores were associated with greater scores on the Coping Motives (CM; drinking to reduce or avoid negative emotional states) subscale of the DMQ, particularly in women. In contrast, ASI scores were not related to scores on either the Enhancement Motives (EM; drinking to facilitate positive emotional states), or Social Motives (SM; drinking to meet social norms) subscales of the DMQ. A regression equation involving scores on the ASI and STAI-T significantly predicted DMQ-CM scores; the regression equation was superior in predicting the frequency of CM drinking for women vs. men. These results imply greater risk for alcohol abuse in individuals (particularly women) who are high in both anxiety-sensitivity and anxiety-proneness.

- T24. Tracy Taylor & Raymond Klein

FIXATION OFFSETS AND COVERT ORIENTING: THE FAILURE TO FIND EFFECTS

When a fixation stimulus is offset prior to saccadic target onset, eye movement latencies are facilitated. The present experiment determined whether facilitation is also observed for covert attentional orienting. Subjects were presented with predictive verbal (endogenous), unpredictable visual (exogenous), or predictive visual (endogenous+exogenous) cues to one of four possible target locations. The target was the appearance of an horizontally- or a vertically-oriented ellipse, in response to which subjects made a speeded manual discrimination. The timecourse of orienting --revealed by the function relating reaction times to cue-target onset asynchrony -- was examined under three "gap" conditions, wherein on one-third of trials there was no fixation offset, on one-third of trials fixation offset occurred simultaneously with cue onset, and on one-third of trials fixation offset occurred 210 msec prior to cue onset. In none of the cue conditions was the speed of covert orienting facilitated by fixation offset.

- T25. Richard Brown, Thane Calder & Charles Gilbert

BEHAVIOURAL DIFFERENCES OF MHC-CONGENIC MICE DURING DEVELOPMENT

Since 1979, it has been known that mice that differ only in the genes of the major histo-compatibility complex (MHC) on chromosome 17 produce different urine odours. These odours thus provide phenotypic markers of the genotype of the mouse which can be used for mate selection and kin recognition. We investigated differences in the development of the behaviour of C57BL/6J (H-2b) and B6.C-H-2bm1 mice, two strains which differ only in the class 1 (H-2K) region of the MHC. Thane Calder found differences in ultrasonic vocalization rates, growth rates, slight movement, grooming and eye opening in these mice and Charles Gilbert found differences in their behaviour in an open field. As developmental biologists have found that MHC-congenic mice differ in their embryonic growth rate, our hypothesis is that postnatal growth and development are also influenced by MHC type. The behavioural differences between genotypes may provide additional information for mate selection and kin recognition.

- T26. Kazuo Semba, Anatomy and Neurobiology

SYNAPTIC CONTROL OF WAKE- AND SLEEP-ACTIVE FOREBRAIN NEURONS

In our continuing effort to elucidate the neural networks involved in the regulation of sleep and wakefulness, we propose a series of working hypotheses that can be used to investigate synaptic control of wake- and sleep-active neurons in the basal forebrain as an integral component of regulatory mechanisms for cortical arousal.

T27. John Barresi

CARGO CULTS, BRANCH DAVIDIANS, BORN AGAIN VEGETARIANS, AND "THE PIOUS GENE."

The present paper is a commentary on a target article to be published in Behavioral and Brain Sciences by D.S. Wilson & E. Sober titled, "Re-introducing Group Selection to the Human Behavioral Sciences." Wilson and Sober argue that the process of group selection has been overly devalued as an evolutionary force during the past thirty years. Using Dawkins' concept of "vehicle", they propose an evolutionary model with a nested hierarchy of vehicles of selection ranging from genes to groups. In my commentary I consider the role that group selection may have played in the evolution of human religious behaviour. In particular, I consider Simon's group selection model of the psychological trait of "docility" and how religious beliefs could serve as absolute authoritative sources for eliciting docile behaviours that are often of genetic, but not necessarily of "rational", self-interest to the individual.

- P1. Heather Schellinck & Richard Brown

SELECTIVE DEPLETION OF GUT BACTERIA ALTERS THE ODOURS OF INDIVIDUALITY IN RATS

Urinary odours of individuality in rats may be produced by the action of specific gut bacteria. To test this hypothesis, we used three types of discrimination tasks to assess the ability of male rats to discriminate between the urine odours of outbred Long-Evans male rats before and after their gram negative or gram positive bacteria were depleted by drug treatment. Results were similar whether gram positive or gram negative bacteria were removed. When tested in an operant-olfactometer, rats previously trained to discriminate between the urine odours of two rats did not generalize to the urine odours of the same rats collected after their bacteria were removed, indicating that removal of the bacteria altered their individual odours. However, these same subjects made fewer errors in learning to discriminate between urine samples from rats after their bacteria were selectively removed than in learning to discriminate between samples from previously unknown rats indicating that they recognized some components of these odours (savings effect). In a habituation-dishabituation task, rats detected discriminable differences between the urine odours of the same rat before and after its bacteria were selectively depleted further indicating that bacterial depletion altered the rat's odour. The fact that the odours of two individual rats could still be discriminated after removal of bacteria indicates that, while their individual odours were altered by removal of bacteria, they were not eliminated.

- P2. Janet L. Ingles, C. Charles Mate-Kole, & John F. Connolly

DISSOCIATION OF VISUAL AND AUDITORY COMPREHENSION IN A CASE OF FLUENT APHASIA

A 24-year old woman developed fluent aphasia and anomia following bitemporal brain damage due to herpes simplex encephalitis. Her language disturbance showed exceptional features: reading was neologistic and yet reading comprehension was superior to auditory comprehension. Event-related brain potential responses (N400) recorded in response to semantically-incongruous sentence endings in the visual but not auditory modality further support this dissociation. These findings suggest that despite a breakdown of phonological output some preservation of visual and semantic routes for reading may occur. These features are discussed with reference to other acquired language disorders.

- P3. Susan Morehouse & John Fisk

ATTENTION: HOW DOES IT MOVE US? A KINEMATIC ANALYSIS OF THE EFFECTS OF ATTENTION ON MANUAL AIMING MOVEMENTS

How does the way we attend to objects influence aspects of our movement to these objects? The effect of attention on the kinematic parameters of motor function has largely been ignored. Since attention and motor function are intertwined in everyday life, it makes sense from an ecological perspective to empirically investigate their interaction. The proposed experiments have been designed to investigate the effects of visually directed attention on the kinematic parameters of manual aiming movements. The paradigm incorporates basic aspects of Posner's (1980) cost/benefit cuing paradigm with those of Rosenbaum's (1980) precuing paradigm for motor function. Possible outcomes of these experiments will be discussed as will potential contributions of this research to the areas of attention and motor function.

- P4. P.A. McMullen, Psychology & T. Trappenberg, Computer Science

REPRESENTING FACES IN A NEURAL NETWORK

A simple perceptron neural network was implemented to represent three distinct faces based on Tanaka's (1991) caricature recognition model. Modifications to his original model were compared including passing the output through a linear transfer function, adding a hidden layer with back propagation, back propagation with momentum and back propagation, with momentum and an entropic error surface and a winner-takes-all network. As was found with the original model, all modified models correctly "recognized" each face input vector and produced higher output activation levels to caricatures than veridical faces. However, only the original model produced equal activation levels in all three output nodes to an average prototypical face. Output nodes of all modified models failed to show any activation to prototypical input. These findings suggest that the system which represents a generic or superordinate face may be independent of the system which represents specific face exemplars.

- P5. G. Rioux

THE APPLICATION OF EVENT-RELATED BRAIN POTENTIAL MEASURES TO NEUROPSYCHOLOGICAL ASSESSMENT

In general, this research is aimed at the development of electro-physiologically-based assessment tools for use with patients who are unable to be tested in the conventional manner on standard neuropsychological tests (e.g., expressive aphasics, coma patients). The present study represents an initial validation of an event-related brain potential (ERP) adaptation of the vocabulary sub-tests of the Wechsler Intelligence Scale for Children, and the Wechsler Adult Intelligence Scale. Preliminary findings with normal subjects are presented, and suggest that a late positive component (i.e., P300) in the ERP waveform may be an accurate index of individual test performance, independent of overt behavioural response.

P6. Laura A. Chambers & Karina Davidson

REDUCING PSYCHOLOGICAL RISK FACTORS: A HEALTH PROFESSIONAL INTERVENTION

The Nova Scotia Heart Health Project was undertaken in the summer of 1989 to enable Nova Scotians to take action to improve cardiovascular health by controlling or reducing their risks for disease. As part of this multifaceted community wide effort, materials aimed at increasing the effectiveness of a psychological intervention will be provided to health professionals. Included in such materials will be a treatment manual providing practical information on how to identify and manage psychological risk factors for CHD such as hostility and social support. For example, information will be provided on how to quickly assess social network and then give practical advice on social network enhancement. Once psychological risk factors have been managed, adherence to interventions for other established risk factors such as obesity and smoking is expected to increase. This practical guide for health professionals will be presented. It will then be used, in addition to other materials, in Nova Scotia to assess the efficacy of reducing community level risk factors.

P7. Alex Trillo

- P8. Farhad N. Dastur & Richard E. Brown

THE EFFECTS OF HYPOTHYROIDISM ON THE ULTRASONIC VOCALIZATIONS OF INFANT LONG-EVANS RATS (*RATTUS NORVEGICUS*)

Eighty Long-Evans rat pups were made hypothyroid by administration of methimazole (MMI) in the drinking water of their dams (e17 - pnd10). Dams were randomly assigned to one of four conditions: (1) control (no drug in water); (2) 0.05 mg/ml (MMI dissolved in water); (3) 0.1 mg/ml, or (4) 0.2 mg/ml. The dependent measure was the number of ultrasonic vocalizations (UVs) emitted during an 8-minute test on postnatal days 8, 10, 12, 14, and 21. Hypothyroidism had a significant effect on UVs, $F(3,76) = 33.40$, $p < .001$, and was significant at all ages tested ($p < .001$). The hypothyroid pups vocalized more than the control pups at all ages except day 8, suggesting a developmental shift towards later ages in the pattern of UV production. There was no effect of sex; however, the age effect was significant, $F(4,304) = 1159.59$, $p < .001$. A significant age x drug interaction was also found, $F(12,304) = 15.61$, $p < .001$. Since the internal body temperature of hypothyroid pups was lower than that of controls, $F(3,72) = 81.90$, $p < .001$, it was concluded that the hypothyroid pups experienced greater thermal stress than controls and this accounted for their increased number of UVs.

- P9. Tara L. Fidler & Vincent M. LoLordo

STRESS AND ALCOHOL PREFERENCE

Volpicelli, Ulm, and Hopson (1990) found that rats given a choice between 5% ethanol and water in the home cage showed enhanced ethanol preference following four daily 1-hr sessions of inescapable shock in another environment, relative to a no-treatment control. We attempted to replicate this effect including a shocked group, a no-treatment control, and a group taken to the distinctive environment each day but not shocked.

Our baseline ethanol preferences were much higher than theirs, 0.55 versus 0.35. Additionally, whereas their ethanol preferences increased from baseline to shock to post-shock phases, our ethanol preferences decreased over phases, especially in the no-treatment group. In the second replication, the pattern of results was the same, although baseline preferences were like those of Volpicelli et al. Thus we did not replicate the stress effect on alcohol consumption observed by Volpicelli et al. Additional research aimed at resolving this discrepancy is in progress.

P10. Judith Francis

DYSPHORIA AND UTILIZATION OF STEREOTYPES

The present study investigated the relationship between dysphoria and utilization of stereotypes. Dysphoric and non-dysphoric subjects were asked to view photographs of male and female target people and asked to guess the height of each target to the nearest half inch. In the no-instruction condition, subjects were simply asked to guess the height of each target. In the instruction condition, subjects were told that male and female targets were actually matched for height so that the stereotype that men are taller than women would not be a useful heuristic to aid them in making their judgements. Stereotype utilization was measured by subtracting the average height estimate of female targets from the average height estimate of male targets. It was hypothesized that dysphoric subjects would rely more heavily on the stereotype that men are taller than women to make their height estimates. However, it was found that dysphoric subjects were actually less likely to rely on the stereotype to make their judgements than non-dysphoric subjects. Results are consistent with research suggesting that dysphoric people are more effortful information processors than non-dysphoric people.

P11. Michael MacGregor & Karina Davidson

GENDER DIFFERENCES IN THE RATING OF HOSTILITY

The effect of coder gender on hostility ratings of structured interviews (SIs) was examined. Two male and two female actors were trained to exhibit high levels of both emotional and behavioural hostility during an SI. The actors were interviewed by either a male or female interviewer, resulting in four SI videotapes representing all possible interviewer/interviewee gender combinations. Trained male and female coders independently rated the resulting SI videotapes to ensure that there was an equal intensity of behavioural and emotional hostility displayed in each videotape. All four videotapes were then rated for hostility level in a counterbalanced order by 105 male and 116 female undergraduates.

There was no main effect for gender of the undergraduate coder or gender of the interviewer in either emotional or behavioural hostility ratings. However, there was a main effect for gender of the interviewee. Female interviewees were rated as being more behaviourally hostile ($F(1,242) = 6.31, p < .01$) and less emotionally hostile ($F(1,242) = 8.42, p < .01$) than male interviewees.

Results are discussed in terms of the possible impact gender may have for future research using SI scored hostility.

- P12. Susan D.Sweet, George R. Rioux & Patrick J. McGrath

PAIN INDUCED CENTRAL PLASTICITY: A REVIEW

This study reviews the literature examining changes pain causes in the central nervous system, which we may call pain induced plasticity. Such studies show that pain may sensitize the body to further pain, or, conversely, that it may inhibit the sensation of further pain. Studies examining pain induced plasticity through self-reports of pain sensitivity or pain threshold by means of the cutaneous flexor reflex were considered for inclusion. Studies were rated by two independent observers on criteria examining their methodology and the type of pain produced. Nineteen studies met the initial criteria and were further examined to determine consistencies which would be useful to create a model of the differential effects pain may have on the central nervous system. Future directions of this work are discussed in terms of its implications for pain assessment and management, especially for children too young to report their level of pain.

- P13. C. A. Dywan, J. McGlone, & A. Fox

UNAWARENESS OF HEMIPLEGIA MAY BE LINKED TO POOR RECOGNITION MEMORY FOLLOWING INTRACAROTID SODIUM AMOBARBITAL INJECTIONS CONTRALATERAL TO THE TEMPORAL LOBE SEIZURE FOCUS

Unawareness of hemispheric deficits experienced following intracarotid barbiturate injections (i.e., Wada testing) has been assessed upon the recovery of function as a model of anosognosia. Unawareness of drug-induced hemiplegia was examined upon clearing of the deficit in 83 patients who were counterbalanced for side of temporal-lobe seizure focus and who underwent left and right sodium amobarbital injections. Unawareness was common (66%) and occurred with equal frequency following left- and right- injections. These findings were not moderated by sex, handedness or speech dominance. Unawareness was demonstrated more frequently after injections contra- than ipsi-lateral to the major seizure focus. Objective memory performance and unawareness were related after contra- but not after ipsi-lateral injections. Thus when assessed following the recovery of function, unawareness of hemiplegia was found to be related to the functional status of the temporal lobe independently of hemispheric laterality.

Department of Psychology

Twentieth Annual In-House Convention

Program Addendum

(If you keep programs for posterity attach this sheet to the full program before storage!)

Program Changes:

T8 (D. E. Mitchell, G. Bateman, C.J. Beaver & P. Ritchie) is a poster (Thursday)

T11 (F. Dastur) is a poster
(Thursday)

P4 (McMullen & Trappenberg) is a talk, replacing T8 in the program

P6 (L. A. Chambers & K. Kavidson) and P12 (S. D. Sweet, G. Rioux & P. J. McGrath) have switched days, so P6 is on Friday and P12 is on Thursday

Program Additions:

One abstract (P7 A. Trillo, C. Daniels & R. Klein) was submitted too late to be in the original program, and one presentation (T. Smith-Crosby) which was inadvertently left off the program has been added as P14. The abstracts for these presentation appear overleaf.

P7. Alex Trillo, Cupido Daniels & Raymond M. Klein

THE PHANTOM CHEST: A STUDY IN RADIOLOGY

Previous research has shown there to be a roughly thirty percent error rate across all specialties in radiology. Research related to reducing this error rate has looked at everything from screening radiology applicants based on their cognitive or perceptual skills to employing the help of computer classifier programs. In our experiment we endeavored to see if there might be a correlation between a radiologist's visual search skills and their performance on a series of chest x-rays. We developed a set of visual search tasks and used fifty chest films, half of which contained abnormalities and half of which were phantoms, in order to assess whether there was a relationship between these two tasks. We found that there was an approximate 30 percent error rate on the films across level of experience and radiological specialty. Additionally, there were some correlations between the film scores the visual scores, but these are not easily interpretable.

P14. Tracy Smith-Crosby

POSSIBLE CONSTRAINTS ON THE CUES INVOLVED IN JOINT VISUAL ATTENTION

Joint visual attention (JVA) which can be defined as simply looking where someone else is looking, has been found to develop in infants in response to the head turn of an experimenter around 10 to 12 months of age. Recent work, however, has shown that infants of only 8 months of age are able to be conditioned to make a JVA-like head turn response if rewarded by an interesting sight located in the experiment's direction of gaze. These findings suggest learning as a possible mechanism through which JVA may develop. However, subsequent research indicates that infants are unable to be conditioned to make a JVA-like head turn response in a direction opposite to the experimenter's head turn cue. Although this finding cannot rule out learning as a mechanism for the development of JVA it certainly does suggest that the cues which elicit JVA in infants may be constrained. The proposed research intends to determine if the cues involved in JVA may be constrained to those containing movement and if JVA is able to be conditioned to as non-social stimuli. Possible implications for mechanisms of JVA are discussed.